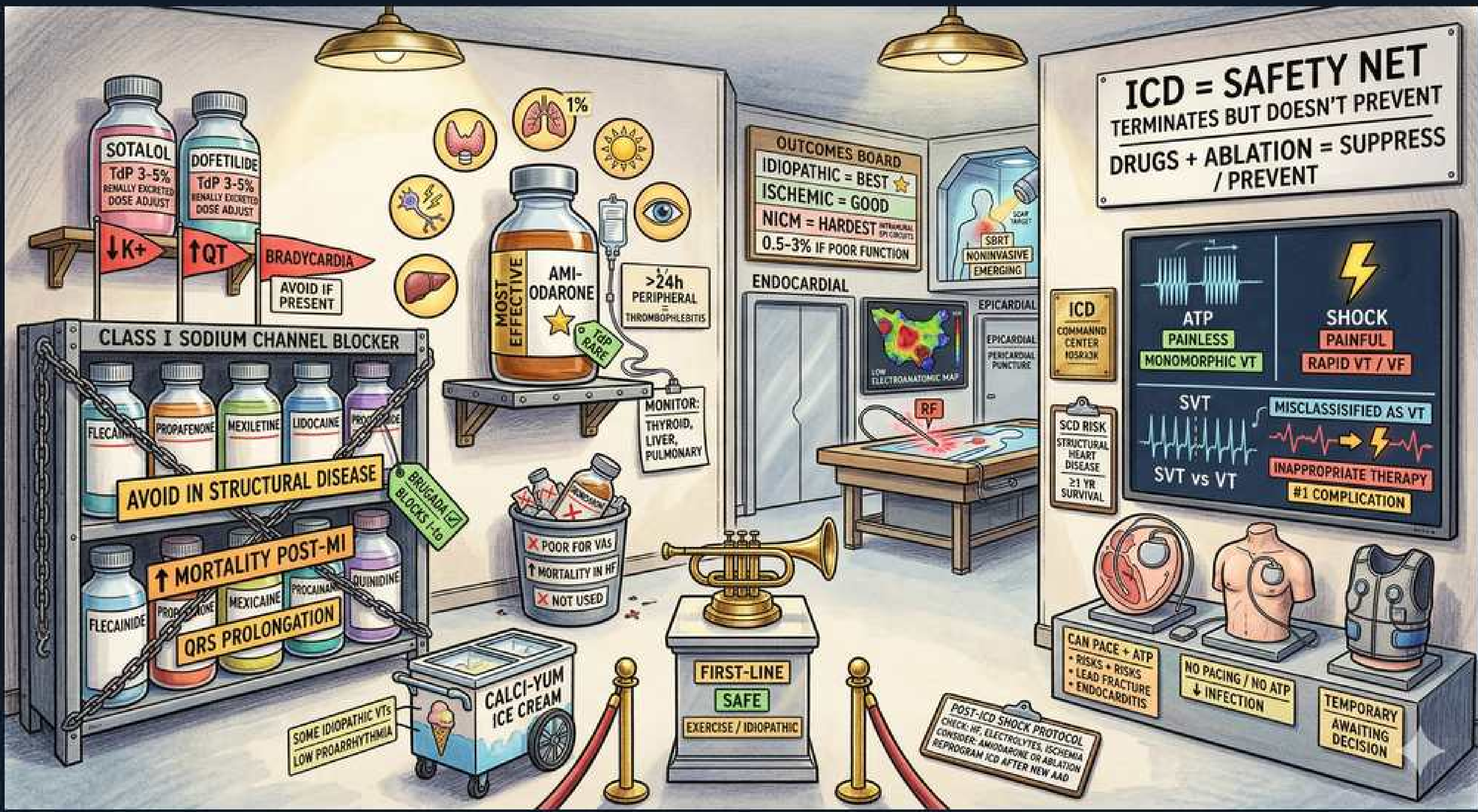
Arrhythmias — Wide-Complex Tachycardia Approach & Treatment



1. SVT vs VT misclassified

WHAT YOU SEE

Sign: 'MISCLASSIFIED AS VT' / SVT vs VT / INAPPROPRIATE THERAPY #1 COMPLICATION

WHAT IT ENCODES

Every regular wide-complex tachycardia (QRS ≥120 ms) must be assumed to be VT until proven otherwise — overall ~80% of WCTs are VT, rising to >95% with prior MI or structural heart disease. Features that PROVE VT: AV dissociation, capture beats (an occasional normal narrow QRS), fusion beats (hybrid QRS), a QRS >140 ms with RBBB shape or >160 ms with LBBB shape, extreme/northwest axis, and positive or negative QRS concordance across V1–V6. Mnemonic: 'old + wide + regular = VT.' Misclassifying VT as SVT-with-aberrancy and giving the wrong drug is the #1 complication.

BOARD TRAP

WRONG: 'A young-looking, hemodynamically stable patient with a wide-complex tachycardia can't have VT, so treat as SVT.' Hemodynamic stability does NOT exclude VT — patients can tolerate VT for hours. The discriminator is the ECG (AV dissociation, capture/fusion beats, concordance) and the history of structural disease, not how 'sick' the patient looks.

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2. ICD = safety net

WHAT YOU SEE

Banner 'ICD = SAFETY NET — TERMINATES BUT DOESN'T PREVENT'

WHAT IT ENCODES

An ICD detects and terminates VT/VF (by anti-tachycardia pacing or shock) but does NOT prevent the arrhythmia from starting — it is the cornerstone of SECONDARY prevention after a survived VT/VF arrest or sustained VT with structural disease, and of PRIMARY prevention when EF ≤35% on optimal therapy. Antiarrhythmics and ablation are layered on to cut the frequency of episodes and shocks.

BOARD TRAP

WRONG: 'Implanting an ICD reduces the number of VT episodes a patient experiences.' The discriminator is mechanism: the ICD only aborts events after they occur and may even deliver shocks for them — to REDUCE the burden of episodes you need antiarrhythmic drugs and/or catheter ablation, not the device itself.

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3. Drugs suppress, ablation prevents

WHAT YOU SEE

'DRUGS + ABLATION = SUPPRESS / PREVENT'

WHAT IT ENCODES

Antiarrhythmic drugs (chiefly amiodarone, sotalol) suppress VT frequency and reduce ICD shocks; catheter ablation targets and eliminates the re-entrant circuit to prevent recurrence. Both are adjuncts to — not substitutes for — the ICD in structural heart disease. Ablation is favored for incessant VT, electrical storm, or when drugs fail or are not tolerated.

BOARD TRAP

WRONG: 'A patient getting frequent appropriate ICD shocks for recurrent VT should have the ICD's detection turned off to stop the shocks.' The discriminator is that the shocks are appropriate (life-saving) — the correct response is to suppress the VT itself with amiodarone/sotalol and/or VT ablation, not to disable therapy.

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4. Class I Na-channel blocker

WHAT YOU SEE

Shelf labeled 'CLASS I SODIUM CHANNEL BLOCKER' (flecainide, propafenone, mexiletine, lidocaine, quinidine), QRS PROLONGATION

WHAT IT ENCODES

Class I antiarrhythmics block fast sodium channels, slowing phase-0 depolarization and prolonging the QRS. Class IA (procainamide, quinidine) and IC (flecainide, propafenone) are the relevant VT agents; class IB (lidocaine, mexiletine) preferentially works on ischemic/depolarized tissue. The key board pitfall is their proarrhythmia in scarred myocardium — slowing conduction in scar promotes re-entry.

BOARD TRAP

WRONG: 'Flecainide or propafenone is a good maintenance antiarrhythmic for VT in a patient with reduced EF.' The discriminator is structure: class IC agents are contraindicated in structural/ischemic heart disease (CAST trial showed increased mortality) and are reserved for AF/SVT in structurally NORMAL hearts.

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5. Avoid in structural disease

WHAT YOU SEE

Green sign 'AVOID IN STRUCTURAL DISEASE'

WHAT IT ENCODES

Class IC agents (flecainide, propafenone) must be AVOIDED in any structural or ischemic heart disease because they increase mortality — the central lesson of the CAST trial, where flecainide/encainide raised death rates in post-MI patients despite suppressing PVCs. The mechanism is use-dependent conduction slowing in scar that facilitates re-entrant VT. Safe-use rule: class IC only with structurally normal hearts.

BOARD TRAP

WRONG: 'Suppressing post-MI PVCs with flecainide will lower the patient's mortality.' This is the exact CAST-trial fallacy — the discriminator is that suppressing ectopy is a surrogate that does NOT equal survival; flecainide INCREASED mortality in this population, so it is contraindicated.

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6. Mortality post-MI

WHAT YOU SEE

Red sign 'MORTALITY POST-MI', POOR FUNC = MORTALITY

WHAT IT ENCODES

Sodium-channel blockers (class IC) increase mortality after MI (CAST), which is the core fact driving drug choice in structural VT: in scarred/ischemic hearts you reach for amiodarone (or sotalol/dofetilide with monitoring), not flecainide/propafenone. Poor ventricular function further worsens prognosis and raises arrhythmic death risk.

BOARD TRAP

WRONG: 'A post-MI patient with reduced EF and frequent VT should be started on oral flecainide for rhythm control.' The discriminator is the post-MI structural substrate — flecainide raises mortality here; amiodarone is the structurally-safe choice, layered onto an ICD.

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7. Amiodarone

WHAT YOU SEE

Jar 'AMIODARONE', AVOID IF / >24h PERIPHERAL line

WHAT IT ENCODES

Amiodarone is the broadly effective, structurally-safe antiarrhythmic for VT — it works in ischemic and non-ischemic cardiomyopathy where class I drugs are dangerous. Acute IV dose for stable VT/VF: 150 mg IV over 10 minutes, may repeat, then 1 mg/min for 6 h and 0.5 mg/min for 18 h (or 300 mg IV push in pulseless VT/VF arrest). Because it causes phlebitis, give via central line if used >24 h. Long-term toxicities: thyroid, pulmonary fibrosis, hepatic, corneal/skin.

BOARD TRAP

WRONG: 'IV amiodarone is the first move for an UNSTABLE wide-complex tachycardia.' The discriminator is hemodynamic status — unstable VT (hypotension, ischemia, shock, altered mental status) demands immediate synchronized cardioversion (or defibrillation if pulseless/VF), with amiodarone only as an adjunct AFTER electrical therapy.

8. Procainamide first-line (stable)

WHAT YOU SEE

'FIRST-LINE' pedestal, EXERCISE / IDIOPATHIC

WHAT IT ENCODES

For STABLE, well-tolerated regular monomorphic VT, IV procainamide is a guideline-preferred first-line agent (PROCAMIO trial showed fewer adverse events and better termination than amiodarone) — dose ~10–17 mg/kg IV at ≤50 mg/min until VT terminates, hypotension occurs, or QRS widens >50%. Amiodarone is the alternative, especially with poor LV function. Stop infusion if QRS widens excessively or hypotension develops.

BOARD TRAP

WRONG: 'Procainamide is the agent of choice for stable VT regardless of ventricular function.' The discriminator is LV function and renal clearance — procainamide is negatively inotropic and can accumulate; in severe LV dysfunction or with active ischemia, amiodarone is generally preferred, and unstable patients still get cardioversion first.

9. ATP — painless

WHAT YOU SEE

'ATP PAINLESS — MONOMORPHIC VT', ICD center

WHAT IT ENCODES

Anti-tachycardia pacing (ATP) is an ICD function that delivers a rapid burst of pacing slightly faster than the VT to penetrate and break the re-entrant circuit, terminating organized MONOMORPHIC VT painlessly and without a shock. It is tried first for slower, organized VT to spare the patient a shock and preserve battery.

BOARD TRAP

WRONG: 'ATP can be used to terminate ventricular fibrillation and very fast polymorphic VT.' The discriminator is organization and rate — ATP only works on organized, monomorphic, re-entrant VT; disorganized VF and very rapid/polymorphic VT require a defibrillation shock, not pacing.

10. Shock — painful, rapid VT/VF

WHAT YOU SEE

Lightning 'SHOCK PAINFUL — RAPID VT / VF'

WHAT IT ENCODES

ICD shock (synchronized cardioversion for organized VT, unsynchronized defibrillation for VF) is the therapy for fast VT and VF and for unstable rhythms. In a coding patient, pulseless VT/VF gets immediate unsynchronized defibrillation; an unstable but perfusing VT gets synchronized cardioversion. Electrical therapy is the priority — drugs are secondary.

BOARD TRAP

WRONG: 'For an unstable wide-complex tachycardia, give a trial of IV amiodarone before attempting cardioversion.' The discriminator is instability — hypotension/ischemia/shock/altered mentation mandates immediate synchronized cardioversion (or defibrillation if pulseless); delaying electrical therapy for a drug trial is the classic dangerous wrong answer.

11. Idiopathic = good outcome

WHAT YOU SEE

Outcomes board 'IDIOPATHIC = BEST', ISCHEMIC = GOOD, NICM = HARDEST, 0.5-3% poor function

WHAT IT ENCODES

Prognosis tracks the substrate: idiopathic VT in a structurally NORMAL heart (fascicular VT, RVOT VT) has the best outcome and is often curable by ablation; ischemic VT has a defined subendocardial target with good ablation success; NICM VT is the hardest to ablate (deep/epicardial scar) with higher recurrence. Idiopathic VTs also respond to specific drugs — verapamil for fascicular VT, beta-blockers/verapamil/adenosine for RVOT VT.

BOARD TRAP

WRONG: 'All VT carries the same poor prognosis and the same ablation success rate.' The discriminator is the substrate — idiopathic VT in a normal heart has an excellent prognosis and high ablation cure rate, whereas NICM VT is the most resistant; lumping them together misjudges both prognosis and the right verapamil-responsive vs amiodarone-requiring therapy.

12. Ablation prevents

WHAT YOU SEE

Ablation catheter / NO PACING figures, TEMPORARY PACING / OVERDRIVE note

WHAT IT ENCODES

Catheter ablation maps and destroys the VT circuit (or scar channels) to prevent recurrence and reduce ICD shocks — it is favored for incessant VT, electrical storm, drug-refractory VT, and idiopathic VT (where it can be curative). Endocardial ablation suffices for subendocardial ischemic scar; epicardial access is often needed for NICM/ARVC. Temporary overdrive pacing can also suppress pause-dependent or incessant arrhythmias acutely.

BOARD TRAP

WRONG: 'A successful VT ablation in a structural-heart-disease patient means the ICD is no longer needed.' The discriminator is residual SCD risk — ablation lowers VT burden and shocks but does not eliminate the risk of future VF, so the ICD stays; ablation is curative only for idiopathic VT in a structurally normal heart.

13. Sotalol / dofetilide shelf

WHAT YOU SEE

Top shelf jars 'SOTALOL' and 'DOFETILIDE' with dosing warnings

WHAT IT ENCODES

Sotalol (class III + non-selective beta-blocker) and dofetilide (pure class III IKr blocker) are alternative antiarrhythmics that suppress VT/AF; both prolong the QT and can cause torsades de pointes, so dofetilide initiation requires inpatient monitoring with renal dose-adjustment, and sotalol needs QT/renal monitoring. Sotalol is an option for VT suppression layered on an ICD when amiodarone is undesirable.

BOARD TRAP

WRONG: 'Sotalol and dofetilide can be started as outpatients without monitoring because they are well tolerated.' The discriminator is QT-driven proarrhythmia — both prolong the QT and risk torsades, so dofetilide legally requires in-hospital telemetry loading and both require renal dosing and QT surveillance.

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14. Outcomes: poor LV function

WHAT YOU SEE

Outcomes board lower line 'POOR FUNCTION' / 0.5-3%

WHAT IT ENCODES

Reduced LV systolic function is the dominant prognostic driver in structural VT — the worse the EF, the higher the arrhythmic mortality and the stronger the ICD indication (primary prevention at EF ≤35% on optimal therapy ≥3 months). EF, not symptom tolerance, drives risk stratification and device decisions.

BOARD TRAP

WRONG: 'A patient with preserved EF and NSVT has the same SCD risk as a patient with EF 25%, so both need an ICD.' The discriminator is EF — primary-prevention ICD hinges on a depressed EF (≤35%); a preserved-EF patient with NSVT generally does not meet primary-prevention criteria absent a high-risk substrate (e.g., LMNA, sarcoid, HCM risk markers).
